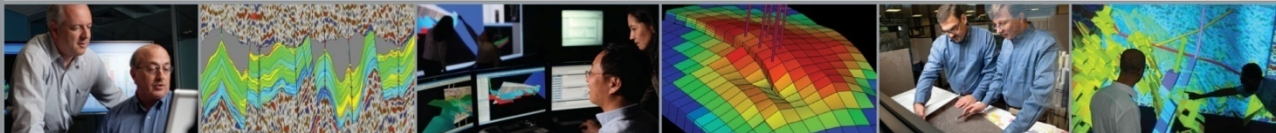


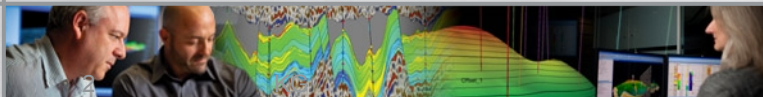
ConocoPhillips Australia Pharos-1 Zero Offset QC VSP Processing

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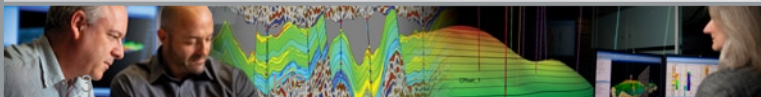
Survey Objectives

Time-Depth relationship (TDR) and velocity information (interval, average and RMS)

Corridor stack for tying to surface seismic

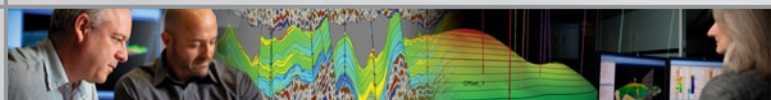
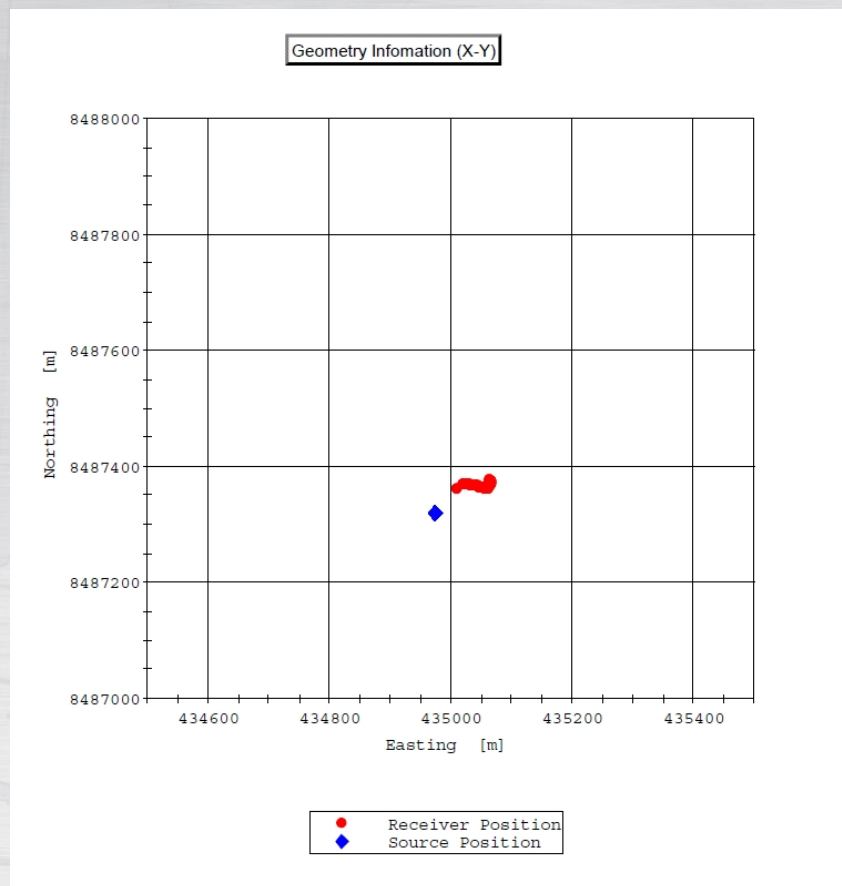
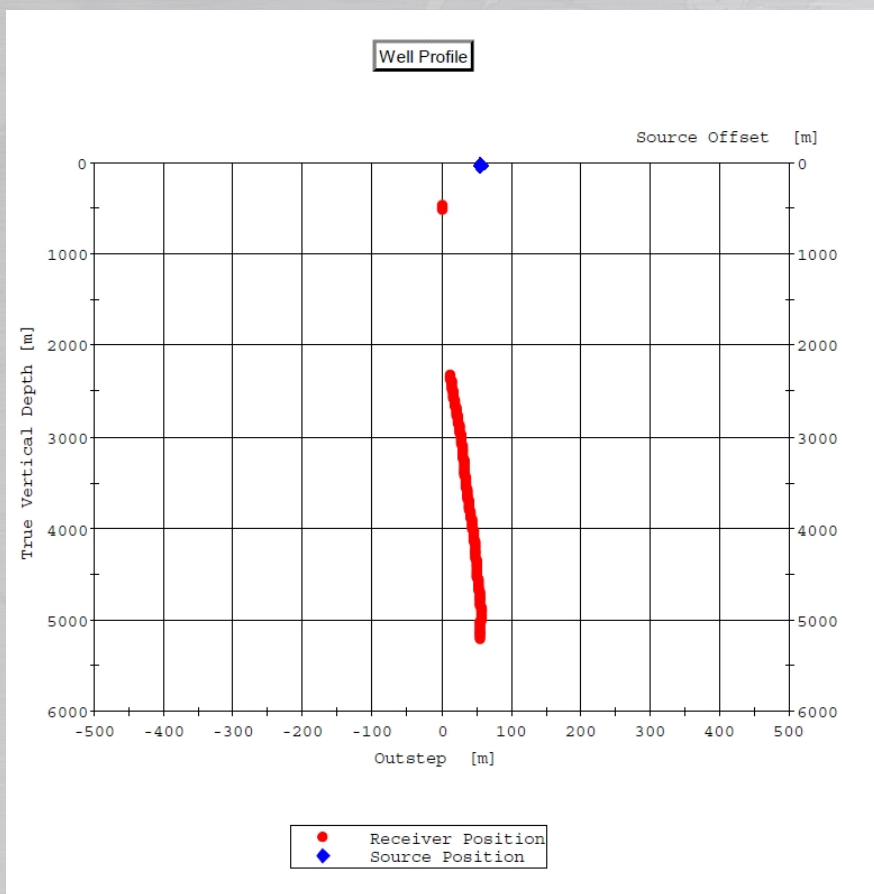
Processing options:

- Sonic calibration and Generation of synthetic seismograms
- Estimation of anellastic attenuation in the vicinity of the well (Q)
- Identification of multiple generators for targeted removal of interbed multiples
- Shear Velocity Analysis



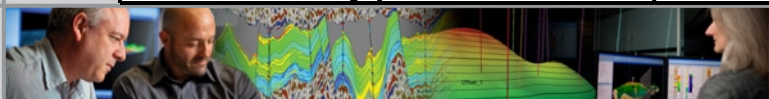
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Pharos-1 Well and Survey Profile

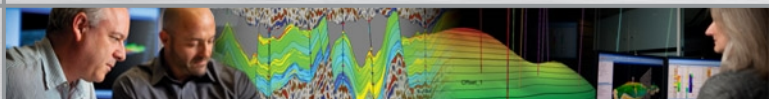
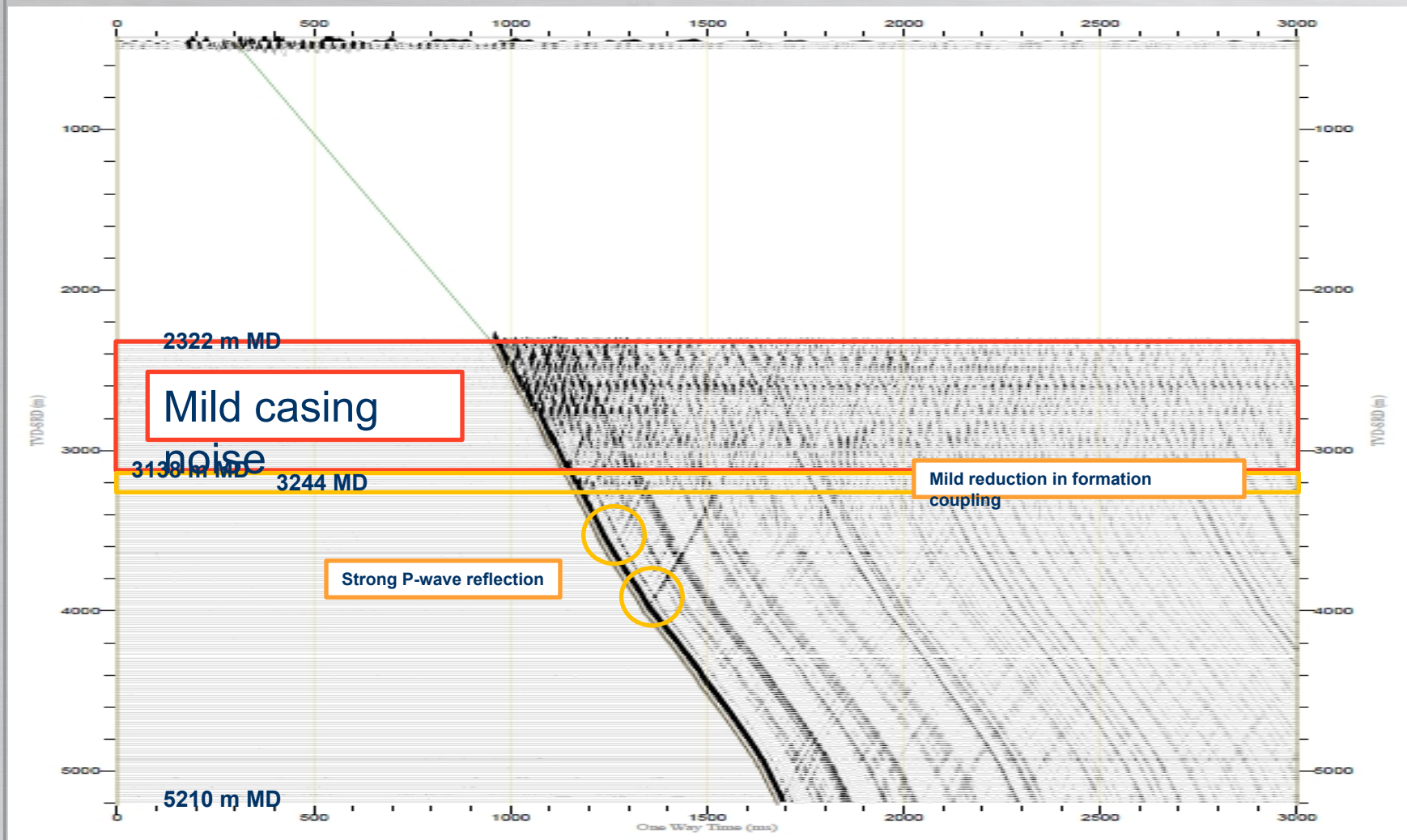


Pharos-1 well survey specifications

Rig	Transocean Legend
SRD	MSL
Elevation of DF	22.1 m above MSL
Elevation of GL	-460 m below MSL
Elevation of SRD	0 m (=MSL)
Well Deviation	max 1.8°
Energy Sources	3×250 cu in G-Gun cluster
Time Reference	FTB Signal from TrisorOFS
Source Offset /	55 @ 220°
Azimuth	5 m / 6.25 m below local tide
Source / Sensor	1524 m/s
Depth	VSI (4 shuttles) / 15.12 m spacing
Reference Velocity	3-C GAC – Geophone Accelerometers
Downhole Tool	
Sensor Type	

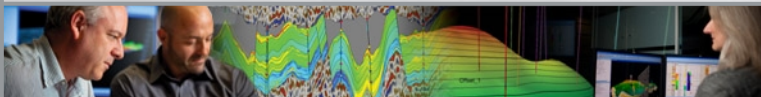
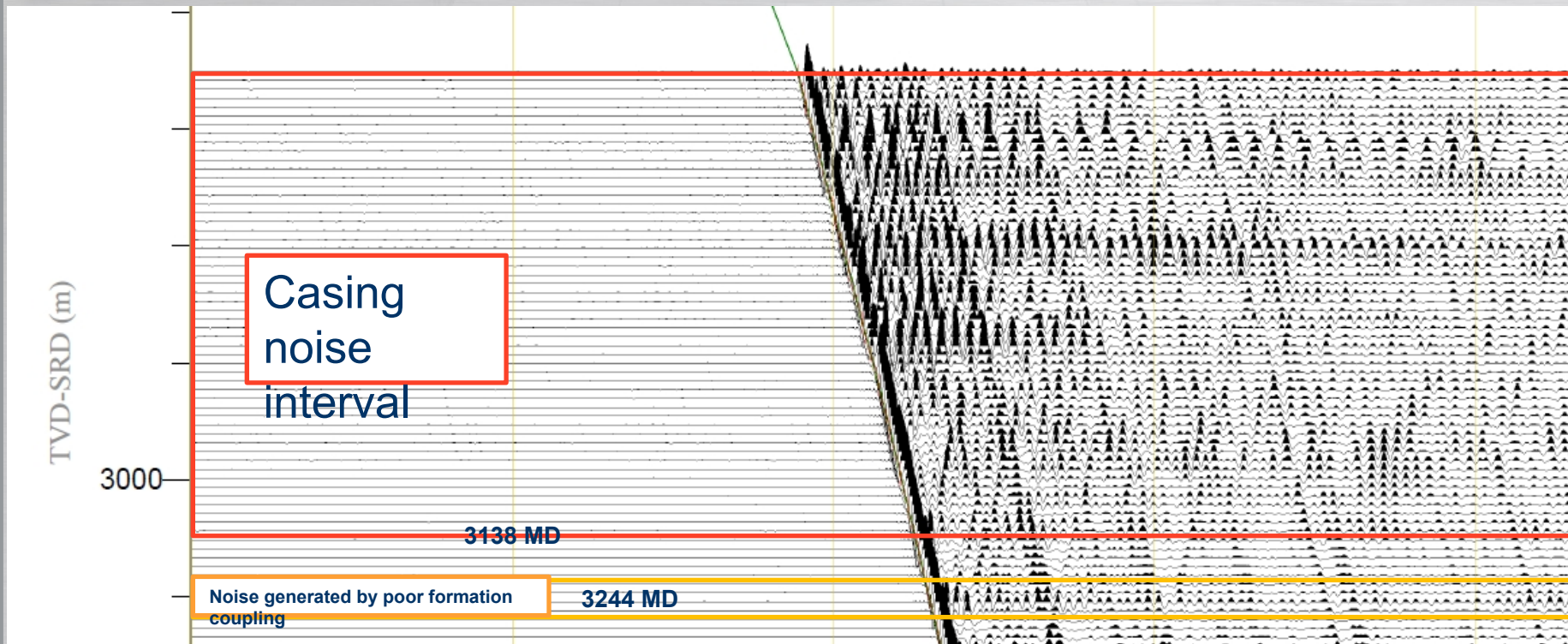


Downhole VSP Data: Vertical Component



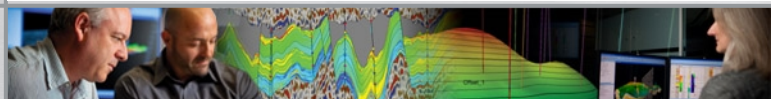
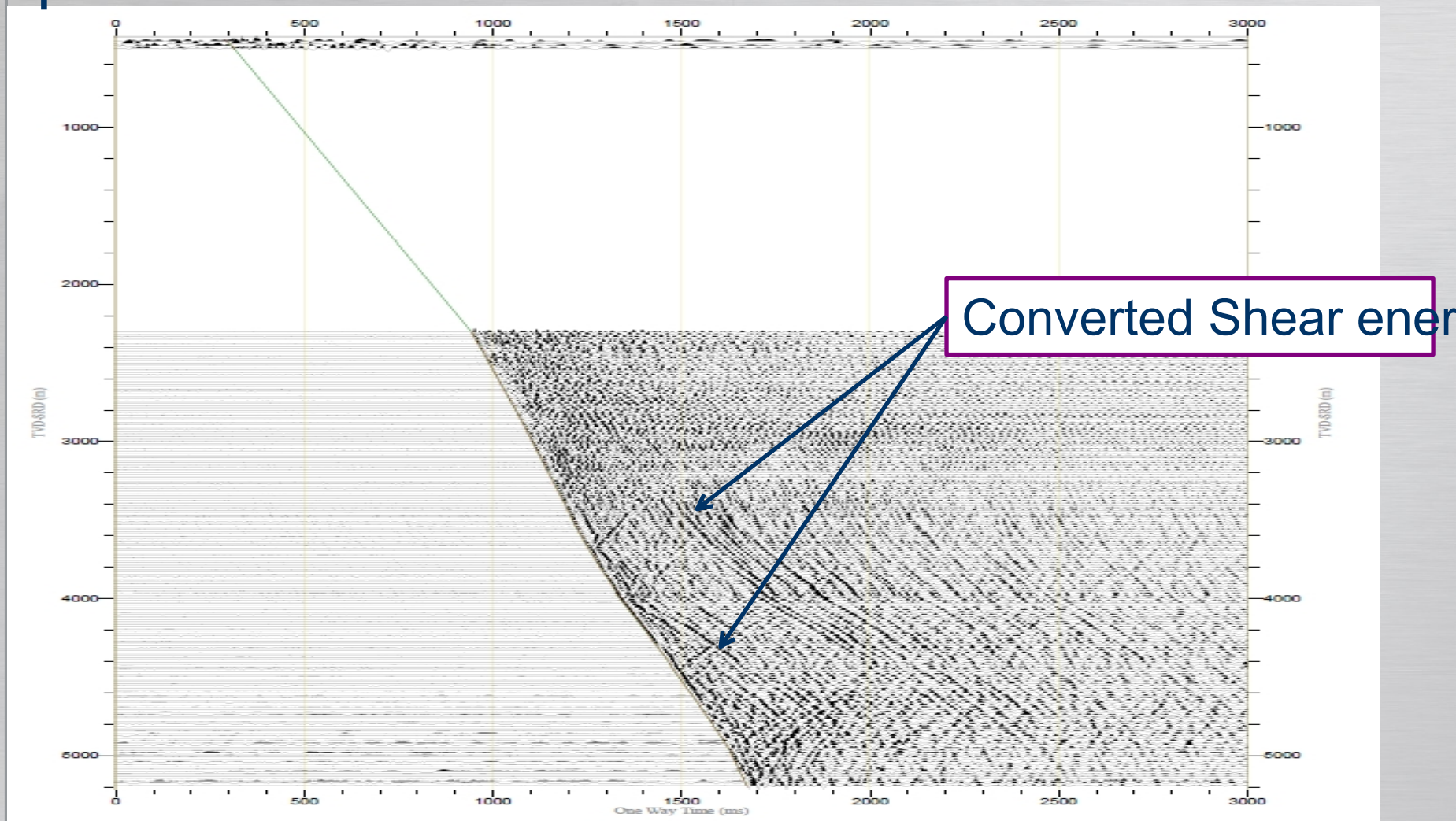
Downhole VSP Data: Noise on Vertical

Casing noise interval is removed in QC processing for corridor stack generation. We can remove casing and poor coupling noise in our full processing and include those intervals in corridor stack.



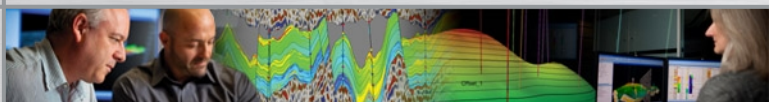
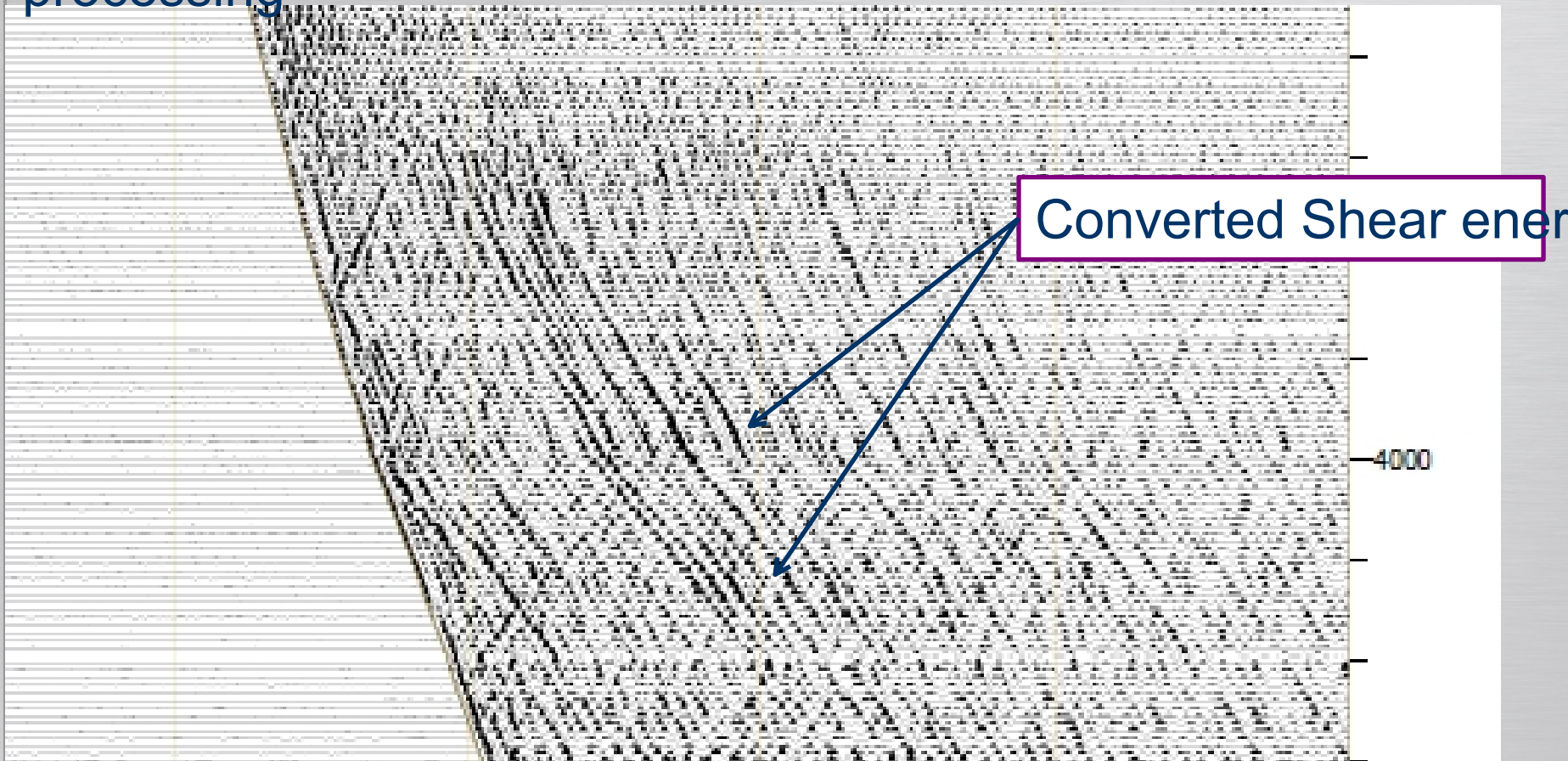
Downhole VSP Data: Horizontal Component

1



Shear velocity estimation in full processing

Shear energy can be utilized to estimate shear velocity in full processing



Downhole VSP Data: Horizontal Component

2

